

Model of a central lateralization processor—Quantitative evaluation

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ear alone) for all types of noise and for the two values of interaural correlation employed. For the briefer (30 ms) stimuli thresholds varied from about 1.3 dB (for wideband, interaurally correlated noise) to about 3 dB (for narrow noise centered at 500 Hz, interaurally uncorrelated). The results will be discussed in terms of the tradability of bandwidth, duration, and interaurally redundant information available in noisy stimuli. [Supported by NSF.]

1:35

QQ3. Lateralization and loudness of high-frequency tones under masking. Mark E. Perkins and M. Pavel (Department of Psychology, New York University, New York, NY 10003)

The apparent location of a 4000-Hz tone presented to both ears will depend on the relative intensity of the signals. The position of the image can be affected by addition of noise in one of the ears; the apparent location will shift toward the ear without noise. This effect of noise on lateralization has some similarity to partial masking. We compared the effect of noise on lateralization and loudness by examining a property of the partial masking data called shift invariance [M. Pavel and G. J. Iverson, *J. Acoust. Soc. Am.* **69**, 1126-1131 (1981)]. Using a transformed up-down method in 2IFC paradigm, subjects adjusted the level of a masked tone in one channel in order to center the image of a tone in quiet presented in the other channel. As expected, these data exhibit "recruitment"-like behavior familiar from studies of partial masking of loudness. The data suggest that shift invariance holds for the lateralization data and therefore provides a convenient means of relating the loudness matching and lateralization tasks.

1:50

QQ4. Model of a central lateralization processor—Quantitative evaluation. Jens Blauert, Werner Lindemann, and Klaus Gruber (Lehrstuhl für allgemeine Elektrotechnik und Akustik, Ruhr-Universität 4630 Bochum, Federal Republic of Germany)

It is established that the hearing system interprets interaural arrival time differences as well as interaural level differences when forming the lateral position of auditory events. Models of binaural signal processing have been constructed which perform the evaluation of arrival time differences and level differences either by means of two separate stages of processing (time difference processor *plus* level difference processor) or by one single stage which can handle time differences as well as level differences. The present paper deals with a model of the latter kind which has previously been proposed by the first author [*Proceedings of the 5th International Congress of Hearing*, edited by v. d. Brink and Bilsen (Delft U. P., Delft, 1980), pp. 421-424]. Examples of a quantitative evaluation of the model with respect to psychoacoustical data on lateralization and lateralization blur will be reported. Further implications of the model will be discussed.

2:05

QQ5. A predictive model for binaural advantages in speech intelligibility. P. M. Zurek (Research Laboratory of Electronics, 36-736, Massachusetts Institute of Technology, Cambridge, MA 02139)

This paper presents a model of binaural advantages in speech intelligibility applicable to anechoic conditions in which speech is masked by a single noise source, with speech and noise sources both located in the horizontal plane. First, frequency- and azimuth-dependent sound-pressure transformations (from Shaw's summary of amplitude measurements and Woodworth's equation for phase) are applied to obtain third-octave signal and noise spectra at each ear. Next (following Levitt and Rabiner), the effective S/N ratio in each band is increased by the masking-level difference, which is approximated with an equation (from Colburn's model of binaural interaction) that describes MLD dependence on relevant stimulus parameters. Mon-

aural and binaural performance are then predicted using Articulation Theory (Kryter). Predictions are in reasonable agreement with data from seventeen studies of the effects of source azimuth and listening mode (monaural or binaural) on intelligibility. The model provides a quantitative framework for analyzing binaural advantages and for assessing the practical benefits of binaural hearing. [Work supported by NIH.]

2:20

QQ6. Binaural horizontal plane localization: Spectral cues as a factor. Alan D. Musicant and Robert A. Butler (Departments of Behavioral Science (Biopsychology) and Surgery, University of Chicago, Chicago, IL 60637).

Binaural localization performance with broadband noise or 4-kHz high-pass noise was investigated under two conditions. An unoccluded pinna condition was compared to performance in a pinna cavity occluded condition. The external meatus was open. Loudspeakers were placed at 15° intervals from 360° through 270° to 180° azimuth. Subjects were asked to identify the loudspeaker from which a sound originated. Results indicated profound disturbances when the pinna cavities were occluded. The occurrence of front-back reversals increased to approximately 40% for all positions with a high-pass noise stimulus. With a broadband noise, only sound sources in the front were transposed to the rear. Accuracy, defined in terms of either proportion correct or error score, decreased significantly ($p < 0.01$) when the pinna cavity was occluded. A paramount role for spectral cues in binaural directional hearing is postulated. [Work supported by NIH.]

2:35

QQ7. Sphere-cylinder discrimination by human observers using broadband sonar pulses. Douglas W. Martin and Whitlow W. L. Au (Naval Ocean Systems Center, Kailua, HI 96734)

Human listeners were presented with target echoes and were asked to indicate which of two target classes, spheres or cylinders, generated the echoes on each trial. Sessions were conducted using either one or two targets from each class. Stimuli were obtained using broadband, ultrasonic, dolphinlike pulses, which were digitally recorded and presented to subjects at 1/50 of the original sample rate. Stimuli thus presented had center frequencies of approximately 2.4 kHz and durations of approximately 50 ms. Targets included spheres and cylinders of various sizes made of solid foam, solid aluminum, and hollow steel. After training, the average probability of correct discrimination with the foam targets varied between 85% and 96% depending on which targets were included in a session. Discrimination between the metal spheres and cylinders always exceeded 90% in both two- and four-target sessions. For both foam and metal targets, subjects reported using two discrimination cues: a higher pitch associated with the cylinder echoes, and more low-frequency reverberation in the sphere echoes. While the target strength of cylinders increases with frequency, and that of spheres does not, the targets could not be separated into classes based only on observing their spectra.

2:50

QQ8. The phase distribution of the pressure alongside a manikin's head and on the surface of a sphere of equivalent perimeter. George F. Kuhn (Vibrasound Research Corp., 10957 E. Bethany Drive, Suite J, Aurora, CO 80014)

The pressure-phase distribution for a frontally incident wave was measured at typical hearing aid locations, alongside the head of KE-MAR and behind its pinna at 0.5, 1.0, 2.0, 4.0, 5.0, 6.3, and 8.0 kHz. The measured phase data and the predictions of the phase on the surface of a rigid sphere of equivalent perimeter as the head are in reasonable agreement throughout, except at 6.3 and 8.0 kHz directly